Homework 3

Microarchitecture Reverse Engineering Across Machines

ECE 492/592 — Fall 2026 | Due November 3, 2026 at 11:59 PM Eastern

> Choose one track

Track A: Individual work without AI. Write all submitted code yourself. Do not submit code found on the internet, including copied or adapted code. Google searches are allowed only for general questions or debugging. Do not use generative AI, including AI-generated search answers, for any part of the assignment.

Strict Track A policy: Using AI or submitting code found online, including copied or adapted code, will result in a failing grade (0/100) for Homework 3.

Track B: Individual work. AI use is permitted, but is neither required nor expected. Using AI carries no penalty. Disclose any AI use. You must understand and explain all submitted work.

> Track A Individual work without AI

Complete all four items without AI (100 points total). Write a paper-style report, as in Homework 1 and Homework 2.

1. Reverse engineer the ROB size. Include your experimental methodology and a code snippet. (40 points)

2. List the challenges and results of your initial naive solutions. Explain step by step how you identified and solved each problem, and show how the results changed. Report any unresolved challenges. (20 points)

3. Measure and compare ROB size across at least eight course machines spanning different processor generations and architectures. (20 points)

4. Sort your results by processor introduction year. Identify a trend or a distinct pattern across architectures, and use your evidence to make and justify a prediction. (20 points)

> Track B Individual work — AI optional

Work individually. The questions below total 300 available points. Choose any combination worth at least 100 points. You only need 100 points for this track; you do not need to complete all questions. List your selected questions and their total points on the first page of your report.

For each selected exercise, run the experiment on two course machines and present the results side by side. Briefly explain your method and any differences between the results. A broader machine sweep, year-sorted trend analysis, and paper-style report are not required for Track B.

1. Design an experiment to confirm whether the processor is pipelined. (10 points)

2. Design an experiment to confirm whether the processor uses out-of-order execution. (10 points)

3. Design an experiment to confirm whether the processor uses branch prediction. (10 points)

4. Reverse engineer the ROB size. (30 points)

5. Reverse engineer the physical register file size. Identify the register class. (35 points)

6. Reverse engineer the number of TLB levels and the size of each level. (35 points)

7. Determine TLB associativity. (15 points)

8. Measure the cost of a TLB miss. (10 points)

9. Compare TLB behavior across supported page sizes. (10 points)

10. Determine how many branches the BTB can store. (20 points)

11. Determine the BTB associativity. (20 points)

12. Investigate which address bits affect BTB indexing and tagging. (20 points)

13. Test whether branch prediction behaves like a neural predictor (such as a perceptron), a counter-based predictor, or a hybrid. Support your conclusion experimentally. (35 points)

14. Estimate the number of instructions fetched per cycle. (10 points)

15. Estimate pipeline depth and explain what your measurement reveals. (10 points)

16. Test whether ROB and BTB resources are shared or partitioned under SMT. (20 points)

Note: Neural predictors can also use tables. Distinguish prediction mechanisms and report uncertainty in your conclusions.

> Requirements for both tracks

Enclave comparison: Run each selected experiment both inside an enclave (for example, Intel SGX) and outside the enclave on the same enclave-capable course machine. Compare the measured results and trends, and explain differences and experimental challenges. Identify the enclave platform and configuration. If enclave support or access is unavailable on a machine, report that limitation; the ordinary experiments must still meet the machine requirement for your track: at least eight for Track A and two per exercise for Track B.

Identify the CPU model and microarchitecture of each tested machine. Track A must also cite each processor’s introduction year. Document unsupported features and access limitations.

For each experiment, include a code snippet in your report and explain the methodology behind it: what you test, how the experiment works, what you measure, and how the results support your conclusion. Support your answers with repeated measurements, plots, and short explanations. Separate experimental findings from published specifications. Cache and prefetcher reverse engineering, including data memory-dependent prefetchers (DMPs), are excluded.

Submit a ZIP containing the PDF report, source code, raw data, plotting scripts, and reproduction instructions. Track A: include sources and a no-AI declaration. Track B: include an AI-use log if AI was used; otherwise state that no AI was used.

> Bonus — Available in both tracks

For any reverse-engineering task, compare your results across at least eight course machines and predict a trend (propose a law). State your proposed law, support it with measurements, and explain its scope and exceptions. In Track A, bonus credit requires analysis beyond the required comparison and trend discussion.